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https://www.tensorflow.org/api_docs/python/tf/gather_nd

Gather slices from params into a Tensor with shape specified by indices.

Function Signatures

```
tf.gather_nd( params, indices, batch_dims=0, name=None )
```

Code Examples

```
tf.gather_nd(  
    params, indices, batch_dims=0, name=None  
)
```

```
tf.gather_nd(  
    params, indices, batch_dims=0, name=None  
)
```

```
tf.gather_nd(  
    indices=[[0, 0],  
            [1, 1]],  
    params = [['a', 'b'],  
              ['c', 'd']]).numpy()  
array([b'a', b'd'], dtype=object)
```

```
tf.gather_nd(  

```

```
indices=[[0, 0],  

```

```
params = [['a', 'b'],  

```

```
['c', 'd']]).numpy()
```

Documentation

Gather slices from params into a Tensor with shape specified by indices.

Used in the notebooks

- Introduction to tensor slicing
 - Parametrized Quantum Circuits for Reinforcement Learning

indices is a Tensor of indices into params. The index vectors are arranged along the last axis of indices.

This is similar to `tf.gather`, in which indices defines slices into the first dimension of params. In `tf.gather_nd`, indices defines slices into the first N dimensions of params, where $N = \text{indices.shape}[-1]$.

Gathering scalars

In the simplest case the vectors in indices index the full rank of params:

In this case the result has 1-axis fewer than indices, and each index vector is replaced by the scalar indexed from params.

In this case the shape relationship is:

If indices has a rank of K, it is helpful to think indices as a (K-1)-dimensional tensor of indices into params.

Gathering slices

If the index vectors do not index the full rank of params then each location in the result contains a slice of params. This example collects rows from a matrix:

Here indices contains [2] index vectors, each with a length of 1.

The index vectors each refer to rows of the params matrix. Each row has a shape of [3] so the output shape is [2, 3].

In this case, the relationship between the shapes is:

It is helpful to think of the results in this case as tensors-of-tensors.

The shape of the outer tensor is set by the leading dimensions of indices.

While the shape of the inner tensors is the shape of a single slice.

Batches

Additionally, both params and indices can have M leading batch dimensions that exactly match. In this case batch_dims must be set to M.

For example, to collect one row from each of a batch of matrices you could set the leading elements of the index vectors to be their location in the batch:

The batch_dims argument lets you omit those leading location dimensions from the index:

This is equivalent to calling a separate gather_nd for each location in the batch dimensions.

More examples

Indexing into a 3-tensor:

The examples below are for the case when only indices have leading extra dimensions. If both 'params' and 'indices' have leading batch dimensions, use the 'batch_dims' parameter to run gather_nd in batch mode.

Batched indexing into a matrix:

Batched slice indexing into a matrix:

Batched indexing into a 3-tensor:

Examples with batched 'params' and 'indices':

See also `tf.gather`.

Returns